Suppose we are given a pair of linear simultaneous equations such as

$a_1x+b_1y+c_1=0$ ...(1)

$a_2x+b_2y+c_2=0$ ...(2)

such that $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. We can solve them as follows :

(1) $\times\ b_2$ – (2) $\times\ b_1$ gives $b_2(a_1x+b_1y+c_1) - b_1(a_2x+b_2y+c_2) = 0$

$$\Rightarrow \quad x(a_1b_2 - a_2b_1) = (b_1c_2 - b_2c_1)$$

$$\Rightarrow \quad x = \frac{(b_1c_2 - b_2c_1)}{(a_1b_2 - a_2b_1)}$$

(1) $\times\ a_2$ – (2) $\times\ a_1$ similarly can be considered and that will simplify to

$$y = \frac{(c_1a_2 - c_2a_1)}{(a_1b_2 - a_2b_1)}$$

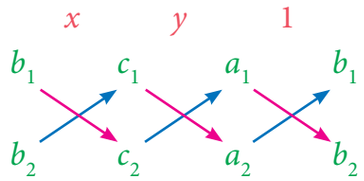
Hence the solution for the system is

$$x = \frac{(b_1c_2 - b_2c_1)}{(a_1b_2 - a_2b_1)}, \quad y = \frac{(c_1a_2 - c_2a_1)}{(a_1b_2 - a_2b_1)}$$

This can also be written as

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

which can be remembered as



Example 3.52

Solve $3x-4y=10$ and $4x+3y=5$ by the method of cross multiplication.

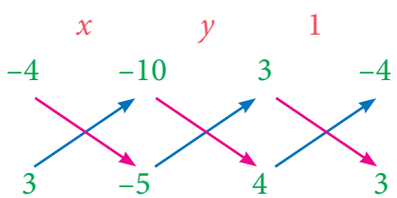
Solution

The given system of equations are

$3x-4y=10 \quad \Rightarrow 3x-4y-10=0$ .....(1)

$4x+3y=5 \quad \Rightarrow 4x+3y-5=0$ .....(2)

For the cross multiplication method, we write the co-efficients as



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